



Schema

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1 Section 1: Core & People

1. Table: Patients

Column Name	Data Type	Constraints	Description
PatientID	INT	PRIMARY KEY, IDENTITY(1,1)	Unique patient identifier
NationalID	VARCHAR(20)	UNIQUE, NOT NULL	National ID number
FirstName	NVARCHAR(50)	NOT NULL	First name
LastName	NVARCHAR(50)	NOT NULL	Last name
Gender	VARCHAR(10)	CHECK (IN ('Male', 'Female')), NOT NULL	Gender
DOB	DATE	NOT NULL	Date of birth
Phone	VARCHAR(15)	NULL	Contact number
Address	NVARCHAR(200)	NULL	Residential address

2. Table: Staff

Column Name	Data Type	Constraints	Description
StaffID	INT	PRIMARY KEY, IDENTITY(1,1)	Staff identifier
FirstName	NVARCHAR(50)	NOT NULL	First name
LastName	NVARCHAR(50)	NOT NULL	Last name
Role	VARCHAR(20)	CHECK (IN ('Doctor', 'Nurse', 'Admin')), NOT NULL	Job role
Specialty	NVARCHAR(100)	NULL	Medical specialty (for doctors)
Phone	VARCHAR(15)	NULL	Contact number

3. Table: Departments

Column Name	Data Type	Constraints	Description
DepartmentID	INT	PRIMARY KEY, IDENTITY(1,1)	Department identifier
DeptName	NVARCHAR(100)	NOT NULL	Department name

2 Section 2: Admissions, Appointments & Beds

4. Table: Appointments

Column Name	Data Type	Constraints	Description
AppointmentID	INT	PRIMARY KEY, IDENTITY(1,1)	Appointment identifier
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), NOT NULL	Patient reference
StaffID	INT	FOREIGN KEY REFERENCES Staff(StaffID), NULL	Doctor reference (for doctor visits)
DepartmentID	INT	FOREIGN KEY REFERENCES Departments(DepartmentID), NULL	Department reference (for para-clinical services)
ApptDate	DATETIME	NOT NULL	Appointment date and time
ApptType	VARCHAR(20)	CHECK (IN ('InPerson', 'Online')), NOT NULL	Appointment type
Status	VARCHAR(20)	CHECK (IN ('Scheduled', 'Cancelled', 'Rescheduled')), NOT NULL	Appointment status
Constraint		CHECK (StaffID IS NOT NULL OR DepartmentID IS NOT NULL)	At least one target must be specified

5. Table: Beds

Column Name	Data Type	Constraints	Description
BedID	INT	PRIMARY KEY, IDENTITY(1,1)	Bed identifier
DepartmentID	INT	FOREIGN KEY REFERENCES Departments(DepartmentID), NOT NULL	Department reference
Status	VARCHAR(20)	CHECK (IN (' Available', 'Reserved', 'Occupied')), NOT NULL	Bed status

6. Table: Inpatient_Transfers

Column Name	Data Type	Constraints	Description
TransferID	INT	PRIMARY KEY, IDENTITY(1,1)	Transfer identifier
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), NOT NULL	Patient reference
BedID	INT	FOREIGN KEY REFERENCES Beds(BedID), NOT NULL	Bed reference
StartDate	DATETIME	NOT NULL	Admission start time on this bed
EndDate	DATETIME	NULL	Discharge or transfer time

3 Section 3: Clinical Records & Lab

7. Table: Medical_Records

Column Name	Data Type	Constraints	Description
RecordID	INT	PRIMARY KEY, IDENTITY(1,1)	Medical record identifier
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), UNIQUE, NOT NULL	Patient reference (1-to-1)

8. Table: Medical_History (Weak Entity)

Column Name	Data Type	Constraints	Description
HistoryID	INT	PRIMARY KEY, IDENTITY(1,1)	History identifier
RecordID	INT	FOREIGN KEY REFERENCES Medical_Records(RecordID), NOT NULL	Medical record reference
ICD_Code	VARCHAR(20)	NOT NULL	Standard ICD disease code
Diagnosis	NVARCHAR(MAX)	NOT NULL	Diagnosis / past disease history
MedicationHistory	NVARCHAR(MAX)	NULL	Patient's medication history
SmokingHistory	VARCHAR(20)	CHECK (IN ('Yes', 'No', 'Former')), NULL	Smoking status
Height_cm	DECIMAL(5,2)	NULL	Anthropometric: height in cm
Weight_kg	DECIMAL(5,2)	NULL	Anthropometric: weight in kg
BloodPressure	VARCHAR(20)	NULL	Vital sign: blood pressure (e.g., 120/80)

RecordDate	DATE	NOT NULL	Date of record entry
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9. Table: Lab_Results

Column Name	Data Type	Constraints	Description
ResultID	INT	PRIMARY KEY, IDENTITY(1,1)	Result identifier
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), NOT NULL	Patient reference
StaffID	INT	FOREIGN KEY REFERENCES Staff(StaffID), NOT NULL	Requesting doctor reference
DepartmentID	INT	FOREIGN KEY REFERENCES Departments(DepartmentID), NOT NULL	Receiving department reference
TestType	NVARCHAR(100)	NOT NULL	Type of test / para-clinical service
ResultDetails	NVARCHAR(MAX)	NULL	Detailed test results (filled when completed)
IsCritical	BIT	NOT NULL DEFAULT 0	Critical result flag for alerts
Status	VARCHAR(20)	CHECK (IN ('Requested','Completed')), NOT NULL DEFAULT 'Requested'	Request status
TestDate	DATE	NOT NULL	Request / completion date

4 Section 4: Inventory, Pharmacy & Prescriptions

10. Table: Inventory_Items

Column Name	Data Type	Constraints	Description
ItemID	INT	PRIMARY KEY, IDENTITY(1,1)	Item identifier
ItemName	NVARCHAR(150)	NOT NULL	Item name
ItemType	VARCHAR(20)	CHECK (IN ('Medicine','Equipment')), NOT NULL	Item type
CurrentStock	INT	NOT NULL DEFAULT 0	Current inventory quantity

11. Table: Inventory_Transactions

Column Name	Data Type	Constraints	Description
TransactionID	INT	PRIMARY KEY, IDENTITY(1,1)	Transaction identifier
ItemID	INT	FOREIGN KEY REFERENCES Inventory_Items(ItemID), NOT NULL	Item reference
TransactionType	VARCHAR(10)	CHECK (IN ('Inbound','Outbound')), NOT NULL	Transaction type
Quantity	INT	NOT NULL	Quantity
TransactionDate	DATETIME	NOT NULL	Exact transaction timestamp

12. Table: Prescriptions

Column Name	Data Type	Constraints	Description
PrescriptionID	INT	PRIMARY KEY, IDENTITY(1,1)	Prescription identifier
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), NOT NULL	Patient reference

StaffID	INT	FOREIGN KEY REFERENCES Staff(StaffID), NOT NULL	Prescribing doctor reference
IssueDate	DATE	NOT NULL	Issue date

13. Table: Prescription_Items (Weak Entity)

Column Name	Data Type	Constraints	Description
PrescriptionItemID	INT	PRIMARY KEY, IDENTITY(1,1)	Prescription item identifier
PrescriptionID	INT	FOREIGN KEY REFERENCES Prescriptions(PrescriptionID), NOT NULL	Prescription reference
ItemID	INT	FOREIGN KEY REFERENCES Inventory_Items(ItemID), NOT NULL	Prescribed item reference
Quantity	INT	NOT NULL	Prescribed quantity

14. Table: Drug_Interactions

Column Name	Data Type	Constraints	Description
InteractionID	INT	PRIMARY KEY, IDENTITY(1,1)	Interaction identifier
DrugA_ID	INT	FOREIGN KEY REFERENCES Inventory_Items(ItemID), NOT NULL	First drug reference
DrugB_ID	INT	FOREIGN KEY REFERENCES Inventory_Items(ItemID), NOT NULL	Second drug reference
Severity	VARCHAR(20)	CHECK (IN ('Critical', 'Moderate')), NOT NULL	Interaction severity
Description	NVARCHAR(MAX)	NULL	Interaction effects description

5 Section 5: Financials & Accounting

15. Table: Invoices

Column Name	Data Type	Constraints	Description
InvoiceID	INT	PRIMARY KEY, IDENTITY(1,1)	Invoice identifier
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), NOT NULL	Patient reference
TotalAmount	DECIMAL(18,2)	NOT NULL	Total service amount
InsuranceShare	DECIMAL(18,2)	NOT NULL	Insurance coverage amount
PatientShare	DECIMAL(18,2)	NOT NULL	Patient payable amount
Status	VARCHAR(20)	CHECK (IN ('Paid', 'Pending')), NOT NULL	Payment status

16. Table: Payments

Column Name	Data Type	Constraints	Description
PaymentID	INT	PRIMARY KEY, IDENTITY(1,1)	Payment identifier
InvoiceID	INT	FOREIGN KEY REFERENCES Invoices(InvoiceID), NOT NULL	Invoice reference
Amount	DECIMAL(18,2)	NOT NULL	Payment amount
PaymentType	VARCHAR(20)	CHECK (IN ('Deposit', 'FullSettlement')), NOT NULL	Payment type
PaymentDate	DATETIME	NOT NULL	Payment timestamp

6 Section 6: IoT & Alert System

17. Table: IoT_Devices

Column Name	Data Type	Constraints	Description
MAC_Address	VARCHAR(17)	PRIMARY KEY	MAC address of device
DeviceType	NVARCHAR(100)	NOT NULL	Device type (vital band, temp sensor)
Status	VARCHAR(20)	CHECK (IN ('Active','Inactive','UnderMaintenance')), NOT NULL	Operational status
PatientID	INT	FOREIGN KEY REFERENCES Patients(PatientID), NULL	Connected patient
BedID	INT	FOREIGN KEY REFERENCES Beds(BedID), NULL	Connected bed

18. Table: IoT_Logs (Weak Entity)

Column Name	Data Type	Constraints	Description
LogID	INT	PRIMARY KEY, IDENTITY(1,1)	Log identifier
MAC_Address	VARCHAR(17)	FOREIGN KEY REFERENCES IoT_Devices(MAC_Address), NOT NULL	Device reference
MetricType	VARCHAR(50)	NOT NULL	Metric type (HR, SpO2, Temp)
MetricValue	FLOAT	NOT NULL	Recorded metric value
Timestamp	DATETIME	NOT NULL	Logging timestamp

19. Table: Alerts (Weak Entity)

Column Name	Data Type	Constraints	Description
AlertID	INT	PRIMARY KEY, IDENTITY(1,1)	Alert identifier
LogID	INT	FOREIGN KEY REFERENCES IoT_Logs(LogID), NULL	IoT log reference (nullable)
ResultID	INT	FOREIGN KEY REFERENCES Lab_Results(ResultID), NULL	Lab result reference (nullable)
Severity	VARCHAR(20)	CHECK (IN ('Moderate', 'Critical')), NOT NULL	Alert severity
Status	VARCHAR(20)	CHECK (IN ('Unchecked', 'Acknowledged', 'Resolved')), NOT NULL	Alert status
GeneratedAt	DATETIME	NOT NULL	Auto-generated timestamp
ResolvedAt	DATETIME	NULL	Resolution timestamp
ResponderID	INT	FOREIGN KEY REFERENCES Staff(StaffID), NULL	Staff who handled the alert